

UNIT 4: Weather Analysis Products

a. Satellite Analysis

Introduction to Satellite Imagery

The goal of this objective will be to show you have the capability to perform a basic satellite analysis. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly identify and label various weather features discussed throughout this objective. These foundational analysis concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

Detection Process

Satellite imagery can detect clouds and weather features between weather reporting stations. Even though all satellite systems view the Earth from orbit they do so in different manners with differing coverage and resolutions. Satellite sensors scan the Earth and atmosphere for electromagnetic radiation (EMR). This includes visible, infrared and microwave radiation. Systems onboard the satellite digitize this EMR data to radio waves and then transmit the radio waves to forecast centers to be processed into different satellite imagery.

Imagery Types

Visible (VIS) Imagery

Visible imagery senses reflected, visible radiation. It is only available during the daytime because it detects the illumination (brightness) of objects. The amount of illumination is dependent upon four factors: the time of day, time of year, latitude, and cloud heights.

Infrared (IR) Imagery

Infrared sensors detect emitted short and long-wave infrared radiation. IR imagery is available 24 hours a day and includes the temperature of Earth's surface, which varies diurnally. During daylight hours land is typically warmer than water, so land will appear darker. Conversely, during nighttime hours water is typically warmer than land, so water will appear darker.

Water Vapor (WV) Imagery

Water vapor imagery measures mid-range infrared radiation. Cold temperatures will appear black, while warm temperatures will appear white. Black and darker gray shades depict dry air. Conversely, white and lighter gray shades are assumed to be moist air. Because WV imagery falls on the infrared spectrum, it is available 24 hours a day.

Interpretation Considerations

When viewing satellite imagery, there are several factors that have an influence on the interpretation of the product. Different imagery types are best suited to identify different weather features and conditions. IR imagery is best utilized to identify high clouds due to their colder temperatures. Visible imagery works best when identifying low clouds due to the water droplets in low clouds having higher albedo than the ice crystals in high clouds. WV imagery best detects

middle and upper-level moisture making it useful for identifying the jet stream and areas of upper-level circulation, especially in areas without clouds.

Exercise / Practice: Types of Imagery

Satellite Interpretation

Non-cloud Features

Surface Features

These features do not move during image animation and are most evident under clear skies. Both visible (VIS) and infrared (IR) imagery use various shades of gray to differentiate between features, whether meteorological or non-meteorological.

Atmospheric Features

Dust and Sand. Areas of dust and sand appear in various shapes and sizes and will have a sharp upstream edge (source region) and a diffuse downstream edge. In visible imagery, they appear light to medium gray while on infrared imagery they are virtually invisible without enhancement. That is because there is little to no temperature contrast.

Smoke and Volcanic Ash. Smoke and volcanic ash is plume-shaped (thin line that broadens out downstream). They have a light gray appearance on VIS imagery and appear differently on IR imagery. When visible, smoke has a dark gray color. Although, smoke is normally invisible due to IR resolution and contamination limitations and a lack of temperature contrast. Volcanic ash is always visible on IR imagery in a light gray to white color. It may also look like cirrus clouds.

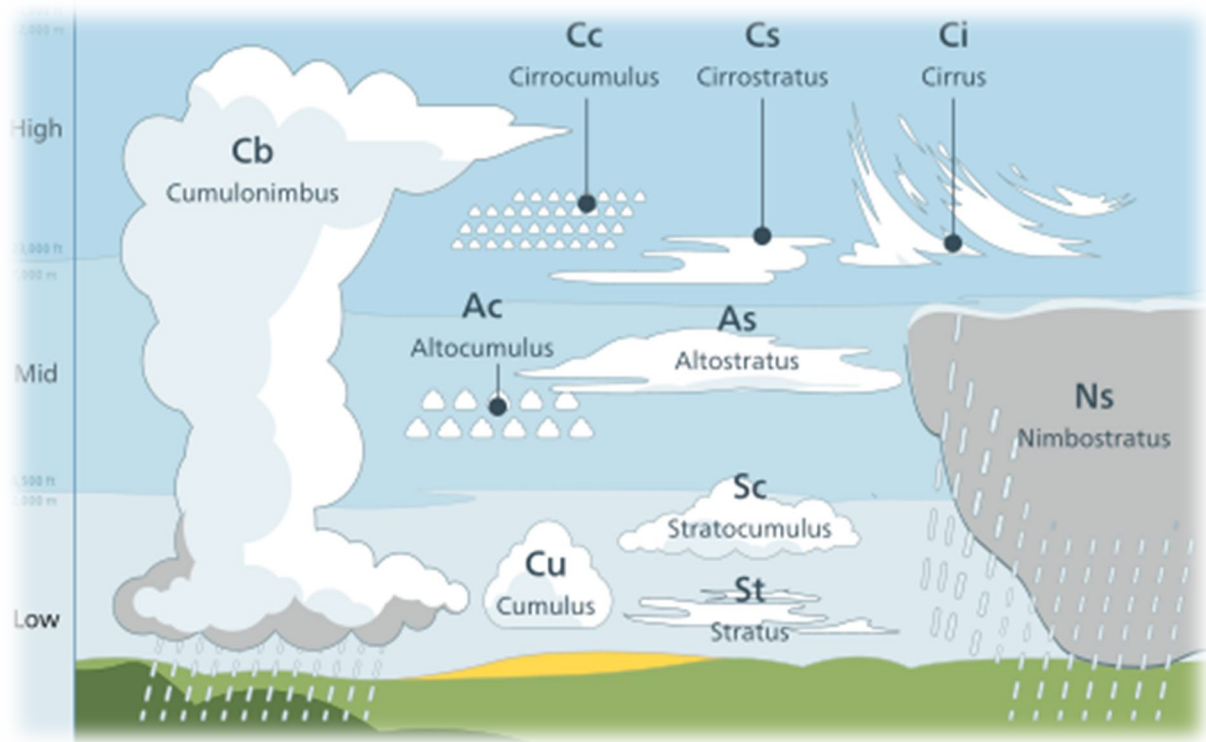
Ship Trails. Anomalous cloud lines may be ship trails formed from a ship's exhaust. They occur most often in persistent maritime anticyclones and may last for days. They are bright white on VIS imagery and virtually invisible on IR due to little or no temperature contrast. If ship trails are on IR, they have a medium gray color.

Contrails. Contrails are small, white cloud lines formed from an aircraft's exhaust. On VIS imagery they may cast a shadow on lower clouds while they are virtually invisible without an enhancement in IR imagery.

Exercise / Practice: Non-Cloud Features

Cloud Features

A cloud is a visible accumulation of minute water droplets and/or ice particles suspended in the atmosphere above the Earth's surface. We separate clouds by the height of the cloud's base using three étages: low, middle, and high.



Low étage – clouds lower than 7,000 feet

Cumulus (CU): Cumulus clouds are small to medium sized with slight vertical extent. They are popcorn-shaped and generally have a bright white appearance.

On VIS imagery, cumulus clouds are small white circular elements in an unorganized cloud field. They generally have a uniform appearance.



Towering cumulus (TCU): Towering cumulus clouds are medium to large with a moderate vertical extent. They resemble columns or towers that are widely spaced apart and their tops and sides are bright white with gray bases.

Towering cumulus also appears as white circular cloud shapes on VIS imagery. The cloud element is like cumulus, but larger. On IR imagery towering cumulus appears lighter gray to white.



Cumulonimbus (CB): Cumulonimbus clouds are large to huge with great vertical extent. Their tops vary in shape; they can be cauliflower or anvil shaped. The tops and sides are bright white with very dark gray to black bases. In some cases, cumulonimbus clouds have pouches hanging underneath the cloud's base. When that occurs, its name changes to Cumulonimbus Mammatus. This is the easiest cloud element to identify.

It has a large circular shape (bigger than towering cumulus), and it is bright white on VIS imagery. It may have an anvil plume with a sharp upstream edge that becomes thin and diffuse downstream. It may also have an overshooting top with small bulges or protrusions centered above the anvil. Cumulonimbus often casts shadows on lower clouds making it easier to detect the tallest cloud tops. On IR imagery, cumulonimbus clouds appear bright white without any enhancement and could have small areas of color indicating the location of overshooting tops.



Fog / Stratus (FG / ST): Fog / Stratus clouds are wide with diffuse edges, and they form in a uniform sheet with no vertical extent. They appear light gray or medium gray color.

Fog and stratus appear the best on VIS imagery. On VIS imagery, fog and stratus appear as a white, uniform sheet. They have sharp outlines without any texture or lumps. There are some differences between the two. For example, fog conforms to terrain features. It also shrinks or

dissipates from the outside edges towards the center. Stratus may move while fog will generally remain stationary. It is best to supplement satellite images with surface observations when it is difficult to tell the difference between them.



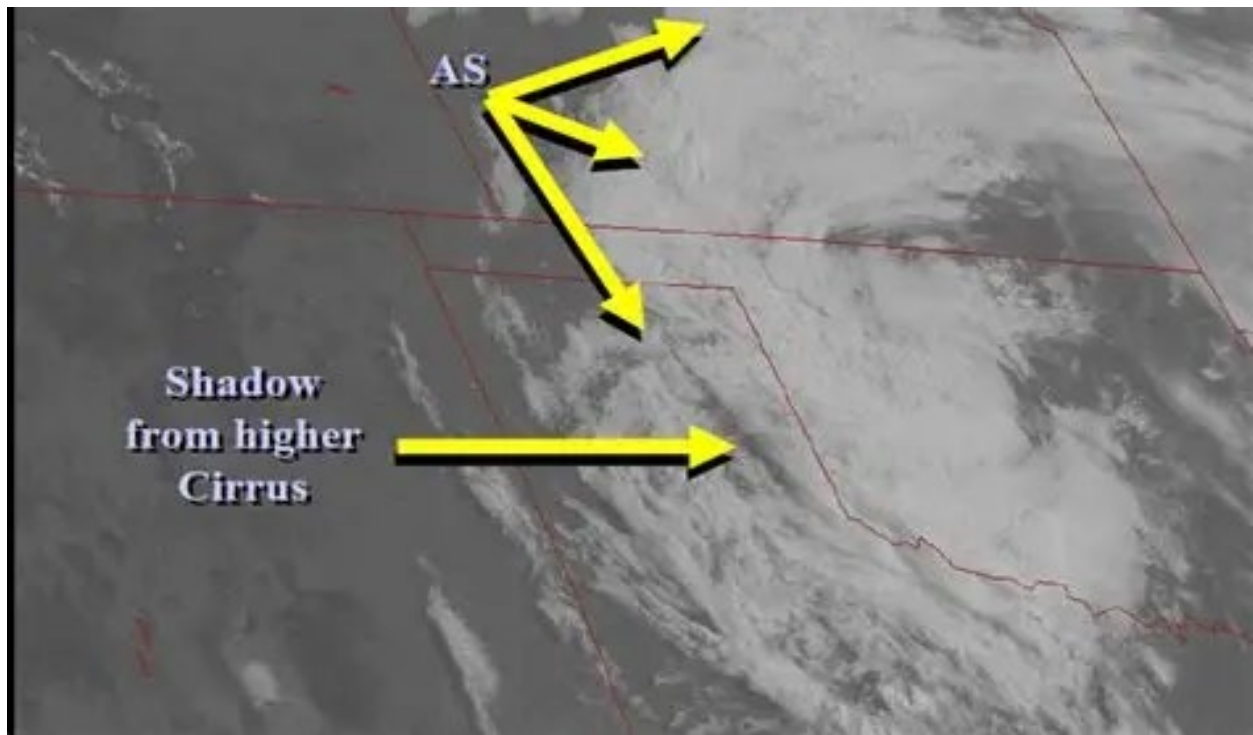
Middle étage - clouds between 7,000 feet and 18,000 feet

Alto cumulus (AC): Alto cumulus clouds resemble stratocumulus clouds, but the individual elements are much smaller. They are widely spaced and circular in shape with little to no vertical extent and are "dirty" white to light gray.



On VIS imagery alto cumulus is a white, non-uniform cloud shield. It has a lumpy texture and individual cloud elements are hard to see. It looks like stratocumulus, but the texture is more visible, and alto cumulus is brighter.

Altostratus (AS): Altostratus clouds form in a continuous, uniform sheet with diffuse edges and they normally cover the entire celestial dome. They have no vertical extent, and their appearance is light gray.



Altostratus and nimbostratus both appear light gray to white on VIS imagery and have a uniform cloud field with little to no texture.



Nimbostratus (NS): Nimbostratus is a thick, dark gray cloud with little to moderate vertical extent. The vertical extent can be misleading since it is different from the other clouds. Nimbostratus forms in the middle étage, but as it develops the cloud's base lowers towards the Earth's surface. Oftentimes it lowers into the low étage.

Nimbostratus appear lighter gray to white on VIS imagery. They have a uniform cloud field with little to no texture.



High étage – clouds higher than 18,000 feet.

Cirrus (CI): Cirrus clouds are thin and wispy forming in hooks or filaments. They have no vertical extent and are bright white.

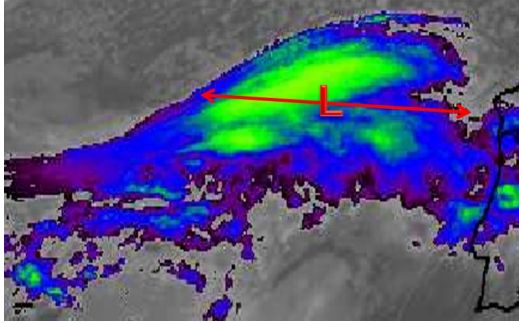
Cirrus has a milky white appearance on VIS imagery. Thin cirrus is usually invisible due to contamination while dense cirrus may cast shadows on lower clouds. Regardless, cirrus forms in streaks or bands. It is bright white on IR imagery. Both thin and dense cirrus are noticeable.



Exercise / Practice: Identifying Cloud Etage

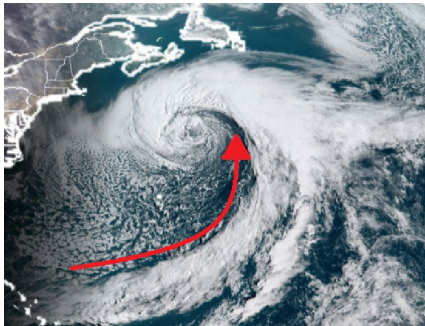
Hemispheric Features

Baroclinic Leaf. Baroclinic leaves are light gray to white on VIS and IR imagery. The bright colors are typically a cirrostratus cloud shield that hides any other cloud elements. The baroclinic leaf normally has a shallow S-shape that may or may not have a V-notch in the trailing edge. The baroclinic leaf typically forms near the base of a long-wave trough.

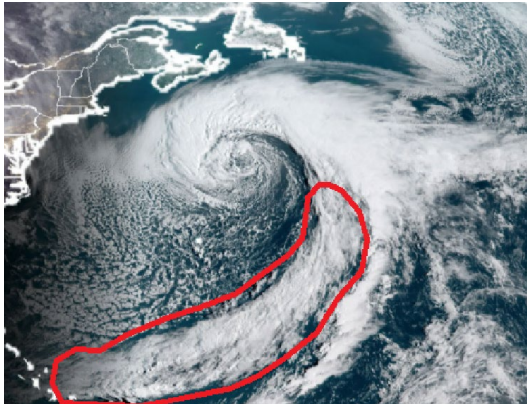


A comma cloud is light gray to bright white on both VIS and IR imagery. Comma clouds will always be found downstream of the long-wave trough axis. There are several parts of the comma cloud.

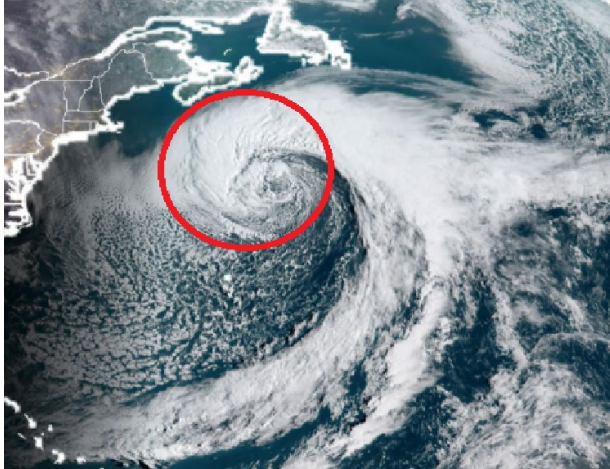
Surge Region: This is the upstream edge of the comma cloud. The surge region is the point where the fastest winds intersect with the comma cloud, typically making it a narrow cloud-free line.



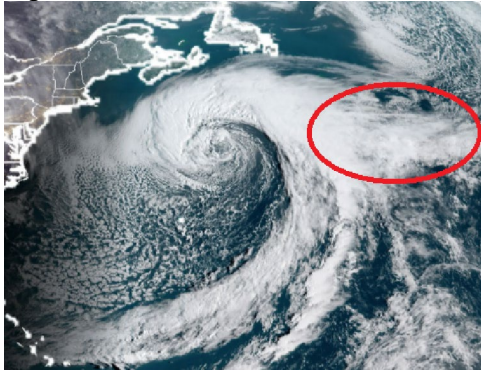
Comma Tail: The comma tail is a long narrow band of cumulonimbus and towering cumulus clouds extending equatorward. The upstream edge is the vorticity comma cloud, and the downstream cirrus shield is the baroclinic zone.



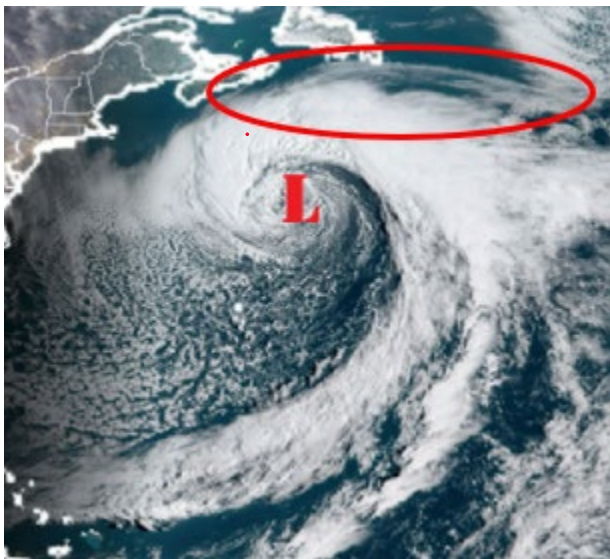
Cusp: The cusp is the upstream edge of the comma cloud. It typically has a cyclonically oriented circular shape.



Baroclinic Zone: The baroclinic zone is in the northeast quadrant of the comma cloud. Best viewed on VIS imagery, it is the area of the bright white cirrostratus before the clouds begin to separate and become more diffuse and fragmented along the leading edge of the comma cloud.



Deformation Zone: The Deformation Zone is the extreme poleward side of the entire comma cloud.



Long-wave trough

The upstream side of a long-wave trough is generally cloud free. Cyclonically orientated, cirrus streaks form parallel to the wind flow outlining a long-wave trough. Analyze the trough axis in the center of the greatest cyclonic turning. Other indicators include a baroclinic leaf and comma cloud. The baroclinic leaf forms near the base of a long-wave trough and comma clouds are always downstream of the long-wave trough axis.

Long-wave ridge

The downstream side of a long-wave ridge is typically cloud free. Anticyclonically orientated, cirrus streaks form parallel to the wind flow outlining a long-wave ridge. Analyze the ridge axis in the center of the greatest anticyclonic turning. Additionally, comma clouds are always upstream of a long-wave ridge axis.

Polar Front Jet

When analyzing for the Polar Front Jet (PFJ) on satellite imagery place the jet axis in the center of a baroclinic leaf's V-notch or nearly center mass of a baroclinic leaf with a well-defined S-shape. It also applies to the center of the U-shape associated with the surge region of a comma cloud. The Polar Front Jet is easiest to identify using Water Vapor imagery by observing the boundary between the dark (dry) and bright (moist) regions in the middle and upper troposphere.

Exercise / Practice: Long-Wave Pattern / PFJ

Synoptic Scale Features

Major short-wave troughs

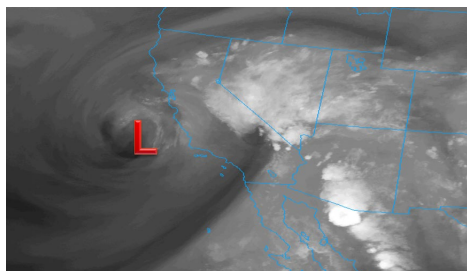
The areas where satellite imagery can help locate major short-wave troughs primarily due to the divergence and upward vertical motion downstream of the trough axis. The first is on the upstream side of TCU or CB clouds. The S-shape of a baroclinic leaf is the second instance. The trough axis is slightly upstream of the cloud mass center.

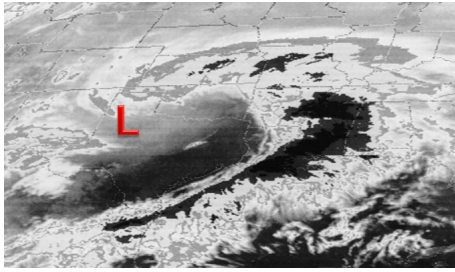
Major short-wave ridges

Short-wave ridges are normally cloud free. Satellite imagery may have anticyclonically oriented cloud elements. A major short-wave ridge may be present there, but it is always best to verify it with another product. Water vapor imagery or animated imagery are best for verifying the location of major short-wave ridges.

Low height centers

In a comma cloud, the low center is located at the cusp. With a cut-off low the image's appearance varies; it can be nearly cloud free or it can have a comma shape. Either way, there will be noticeable cyclonic rotation in the cloud field.





High height centers

High height centers are normally cloud free. There may be a C-shape that appears on the western edge of the high center. The best image for identifying high height centers and the C-shape is water vapor imagery.

Exercise / Practice: Upper Level Features

Low Pressure Center

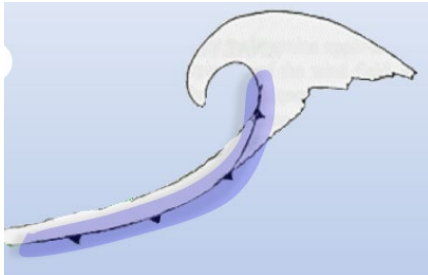
Satellite images display non-occluded and occluded low-pressure systems. They have large cloud masses that typically obscure the low center.

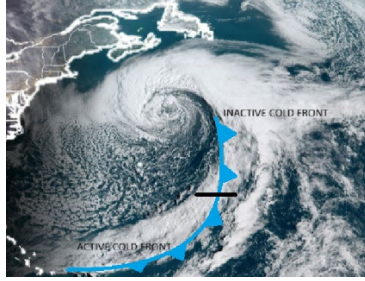
Non-occluded system

A baroclinic leaf represents a non-occluded low-pressure system also known as an unstable wave. The low center is slightly downstream of the baroclinic leaf's center between the cloud's S-inflection point and its easternmost (leading) edge. A cold front is on the equatorward side of the cloud mass extending from the low center to the trailing edge of the S-shape. A warm front is also on the equatorward side of the cloud mass extending from the low center to the leading edge of the S-shape.

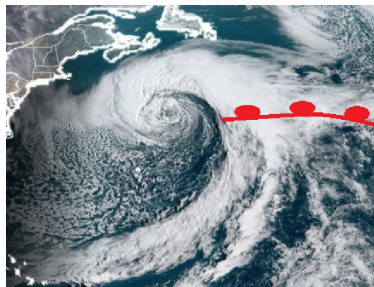
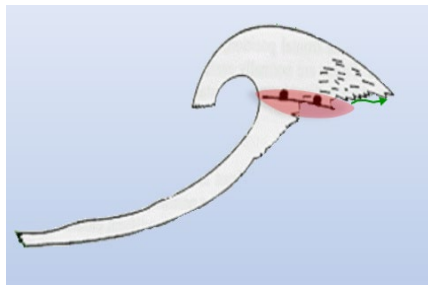
Occluded system

Cold fronts transition through the comma tail extending from the triple point to the end of the comma tail. The inactive cold front begins at the triple point behind the comma tail. It changes to an active cold front where the comma tail bends the most. The active front then extends to the front of the comma tail before ending at the end of the comma cloud.



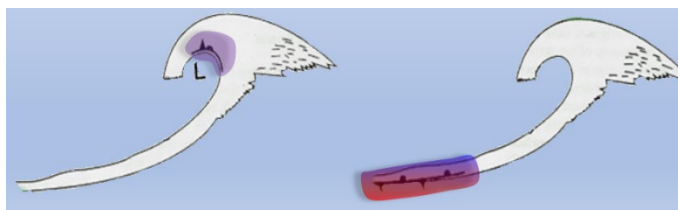


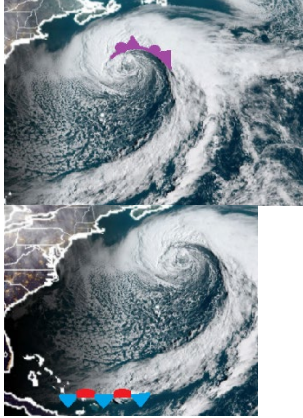
The baroclinic zone often obscures the warm front. The front extends downstream from the triple point to the farthest eastern portion of the cloud shield. If present, the warm front ends at a V-notch or wedge in the leading edge of the baroclinic zone.



Comma clouds are associated with occluded systems whether it is a mature or decaying wave. The low-pressure center location is between the surge region and the cusp for both systems. For mature waves, it is closer to the surge region since the system is still baroclinic. Decaying waves are barotropic, so the low center is closer to the cusp.

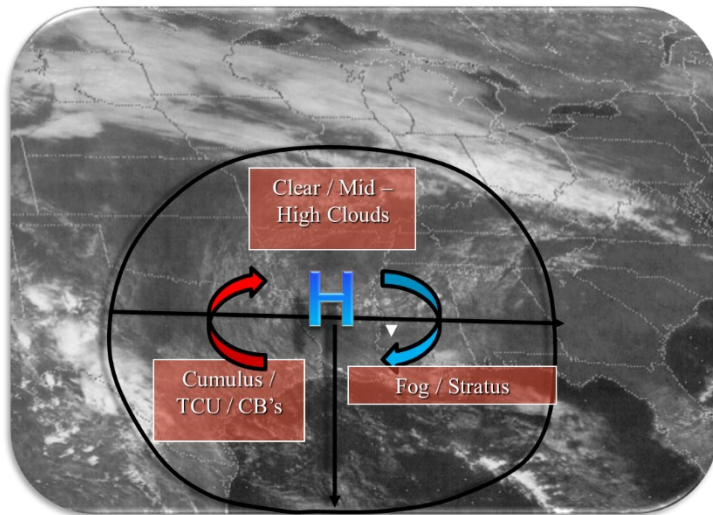
The occluded front extends upstream from the triple point along the bottom edge of the comma head. Stationary fronts are the most difficult to locate. They are on the equatorward side of a fragmented comma tail. The front extends upstream from an active cold front and may continue along the center of a rope cloud if present. Rope clouds are narrow lines of cumulus or towering cumulus clouds that form over water.





High pressure centers

The downward vertical motion associated with a high-pressure center dissipates any clouds making it difficult to locate the center. However, if clouds are present, you will most likely see Cumulus, Towering Cumulus, and Cumulonimbus clouds southwest of the high, and fog or stratus to the southeast of the high. Poleward of the high will usually be clear or have thin mid to high level clouds.



Practice: Satellite
PC: Satellite